

CSIR-UGC NATIONAL ELIGIBILITY TEST (NET)

MATHEMATICAL SCIENCES

Complete Subject & Sub-topic Checklist



Based on complete analysis of

December 2025 & June 2025 Papers

All 240 questions classified | 12 subjects | 100+ sub-topics



Part A (General Aptitude) | Part B (MCQ) | Part C (MSQ) | Code 704 Shift 2

How to use this checklist: Use the ☐ boxes to tick off sub-topics as you study each one. Colour bands indicate topic priority: subjects 1–3 are Tier 1 (highest ROI), 4–6 are Tier 2, 7–12 are Tier 3 / selective. Each subject header shows the question count across both 2025 papers and the marking scheme.

#	Subject	Qs (B+C)	Tier	Part B	Part C
1	Real Analysis	38	1	4–5	8–9
2	Linear Algebra	32	1	4	6
3	Statistics & Probability	30	1	5	8
4	Complex Analysis	26	2	4	5–6
5	Abstract Algebra	24	2	4	5
6	Ordinary Differential Equations	22	2	3	4
7	Partial Differential Equations	18	3	2–3	4
8	Topology	16	3	2	4
9	Functional Analysis	12	3	1	3
10	Calculus of Variations & Mechanics	11	3	2	3
11	Numerical Methods	8	3	2	2
12	Linear Programming & OR	3	4	1	1
A	General Aptitude (Part A)	20/paper	–	–	–

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Real Analysis 38 Questions across both papers | Part B: 4–5 | Part C: 8–9 | **Tier 1 — Highest Priority**

Sub-topics

- **Sequences of Real Numbers**
 - ☐ Convergence and divergence; Cauchy sequences
 - ☐ $\limsup$ and $\liminf$: definitions, properties, squeeze theorem
 - ☐ Monotone sequences; bounded sequences (Bolzano-Weierstrass)
 - ☐ Subsequences and accumulation points
 - ☐ Decimal expansions of real numbers; digit-by-digit locking property
 - ☐ Sequences defined by recursion: $x_{n+1} = f(x_n)$; convergence to fixed points
- **Series of Real Numbers**
 - ☐ Convergence tests: Ratio, Root (Cauchy-Hadamard), Comparison, Limit Comparison
 - ☐ Alternating series (Leibniz test); absolute vs. conditional convergence
 - ☐ Rearrangements of series; Riemann rearrangement theorem
 - ☐ Power series: radius of convergence; term-by-term differentiation/integration
 - ☐ Double series and Cauchy product
 - ☐ Series with log factors: $\sum \frac{1}{n(\log n)^\alpha (\log \log n)^\beta}$ (Bertrand-De Morgan)
- **Limits and Continuity**
 - ☐ ε - δ definition; one-sided limits; limits at infinity
 - ☐ Uniform continuity: definition, characterisation on compact sets
 - ☐ Composition of uniformly continuous functions (e.g., $f \circ g$ when g is x^2 on $(0, 1]$)
 - ☐ Intermediate value theorem; fixed-point theorem
 - ☐ Lipschitz functions; relation to uniform continuity
- **Differentiability**
 - ☐ Mean Value Theorem, Cauchy MVT, Taylor's theorem with remainder
 - ☐ L'Hôpital's rule; indeterminate forms
 - ☐ Functions with prescribed derivatives at 0: $f^{(n)}(0) = n!n^n$ type
 - ☐ Implicit Function Theorem in $\mathbb{R}^n$: existence, uniqueness, derivative formula
 - ☐ Functions of two variables: partial derivatives, differentiability, chain rule
 - ☐ Limits in $\mathbb{R}^2$: path-dependent vs. genuine limits; e.g., $(ax + by)^2 / (ax^2 + by^2)$
- **Riemann Integration**
 - ☐ Riemann integrability: characterisation (Lebesgue criterion — measure of discontinuities is 0)
 - ☐ Fundamental theorem of calculus (both parts)
 - ☐ Improper integrals: convergence on $[0, \infty)$; comparison test
 - ☐ Functions $f \in \mathcal{F}$ (Riemann integrable on $[0, t]$): if f uniformly continuous and $\int_0^\infty f$ converges $\Rightarrow f(x) \rightarrow 0$
 - ☐ Existence of f with convergent $\int_0^\infty f$ but $f \not\rightarrow 0$; oscillating counterexamples
 - ☐ Monotone functions: one-sided limits always exist; discontinuities are countable
 - ☐ Riemann-Stieltjes integration (basics)
- **Sequences and Series of Functions**
 - ☐ Pointwise vs. uniform convergence: definitions, examples, counterexamples
 - ☐ Weierstrass M-test for uniform convergence of series
 - ☐ Continuity under uniform convergence; integration/differentiation term-by-term
 - ☐ Dini's theorem: monotone pointwise $\rightarrow$ uniform on compact set

- ☐ $f_n \rightarrow 0$ pointwise but $\int f_n \not\rightarrow 0$ examples (tent functions)
- ☐ Uniform convergence on sub-intervals; behaviour at boundary
- **Metric Spaces**
 - ☐ Metric space axioms; equivalent metrics; discrete metric
 - ☐ Open and closed sets; interior, closure, boundary
 - ☐ Compactness: Heine-Borel, sequential compactness; finite subcover property
 - ☐ Completeness; Cauchy sequences in metric spaces; Baire category theorem
 - ☐ Distance function $d(x, A) = \inf_{a \in A} d(x, a)$: Lipschitz continuity (hence uniform continuity)
 - ☐ Closed set characterisation: A closed $\Leftrightarrow d(x, A) > 0$ for all $x \notin A$
 - ☐ Compact metric spaces: continuous image is compact; min/max attained
 - ☐ Contraction mapping principle (Banach Fixed Point Theorem)
- **Countability and Set Theory**
 - ☐ Countable and uncountable sets; Cantor-diagonal argument
 - ☐ Decimal expansions with restricted digits (e.g., digits from $\{0, 2, 4\}$): uncountability
 - ☐ Continuous injections preserve cardinality; $f(\mathbb{N})$ vs. $f(\mathbb{R})$
 - ☐ Finite subsets of $\mathbb{Z}$: countable; power set of $\mathbb{N}$: uncountable
- **Monotone Functions and Bounded Variation**
 - ☐ Monotone increasing on $[0, 1]$: one-sided limits exist everywhere; at most countably many discontinuities
 - ☐ Surjective monotone on $[0, 1] \Rightarrow$ continuous; Riemann integrable monotone
 - ☐ $U_\alpha = (L_\alpha^-, L_\alpha^+)$ intervals for distinct α, β are disjoint
 - ☐ Functions of bounded variation; Jordan decomposition

2

Linear Algebra 32 Questions across both papers | Part B: 4 | Part C: 6 | **Tier 1**
— Highest Priority

Sub-topics

- **Vector Spaces and Subspaces**
 - ☐ Subspaces, spanning sets, linear independence
 - ☐ Basis and dimension; Steinitz exchange lemma
 - ☐ Dimension of sum and intersection: $\dim(U + V) = \dim U + \dim V - \dim(U \cap V)$
 - ☐ Quotient space V/W ; dimension = $\dim V - \dim W$
 - ☐ Spanning set $\{u, v, w, x, y\}$ with $\{u, v, x, y\}$ LI: $\dim V \in \{4, 5\}$ argument
- **Linear Maps and Matrices**
 - ☐ Rank-nullity theorem; kernel and image
 - ☐ Matrix representation; change of basis
 - ☐ Invertibility, determinant, adjugate
 - ☐ Trace: $\text{tr}(AB) = \text{tr}(BA)$; trace and eigenvalue sum
 - ☐ Systems of linear equations; row reduction
- **Eigenvalues and Eigenvectors**
 - ☐ Characteristic polynomial; Cayley-Hamilton theorem
 - ☐ Newton's power-sum identities: $\text{tr}(A) = \sum \lambda_i$, $\text{tr}(A^2) = \sum \lambda_i^2$; char poly from trace data
 - ☐ Algebraic and geometric multiplicity; defective matrices
 - ☐ Eigenvalues of functions: $f(A)$ has eigenvalues $f(\lambda_i)$; Cayley-Hamilton application
 - ☐ Largest real eigenvalue of a specific 4×4 matrix (e.g., sum of rows trick)
 - ☐ Matrices over $\mathbb{C}$: trace 1, det 1, one eigenvalue = 1 $\Rightarrow$ structure of other two
- **Diagonalisation and Jordan Normal Form**
 - ☐ Diagonalisability criteria: n distinct eigenvalues; geom. mult. = alg. mult.
 - ☐ If T^2 is diagonalisable, when is T diagonalisable? (need non-negative eigenvalues of T^2)
 - ☐ If T^2 has only negative eigenvalues $\Rightarrow \dim V$ is even
 - ☐ Jordan blocks; minimal polynomial computation
 - ☐ Simultaneous diagonalisation of commuting matrices
- **Inner Product Spaces**
 - ☐ Inner product axioms; Cauchy-Schwarz; triangle inequality
 - ☐ Gram-Schmidt orthogonalisation; orthonormal bases
 - ☐ Orthogonal complement; projection theorem; best approximation
 - ☐ Adjoint of a linear map; self-adjoint, unitary, normal operators
 - ☐ $M_n(\mathbb{C})$ with $\langle A, B \rangle = \text{tr}(A^*B)$; rank- k conditions; $W_k = \{0\}$ for $k \geq 1$
 - ☐ Subspace of $C[0, \pi]$ spanned by $\{f_1, f_2, f_3\}$ with inner product $\frac{2}{\pi} \int fg \sin^2 x dx$
 - ☐ FALSE: "eigenvectors of B for distinct eigenvalues are B -orthogonal" (not true with non-standard inner product $\langle v, w \rangle = v^\top Bw$)
- **Quadratic Forms**
 - ☐ Positive/negative definite, semi-definite, indefinite; Sylvester's criterion
 - ☐ Sylvester's law of inertia; signature; congruence
 - ☐ Quadratic form $Q(f) = \int_{-1}^1 (f')^2 dt$ on $V = \{ax^3 + bx^2 + cx\}$: FALSE statements
 - ☐ Complex quadratic forms over $\mathbb{Q}$; zeros over $\mathbb{C}$ vs $\mathbb{R}$ vs $\mathbb{Q}$
- **Projections and Special Operators**
 - ☐ Idempotent operators: $T^2 = T$; direct sum decomposition $V = \ker T \oplus \text{im } T$
 - ☐ $T^2 = T$: $W = \text{span}\{I, T^n : n \geq 1\}$ has $\dim W = 2$; idempotents in W

- ☐ If $U^2 = U$ and $(T - U)^2 = T - U \Rightarrow (TU)^2 = TU$
- ☐ $A^2 = I$, $\text{Tr}(A) = 0$: anticommutant $\{B : AB + BA = 0\}$ is 8-dimensional $\mathbb{C}$ -vector space
- ☐ Commutant $\{B : AB = BA\}$ also 8-dimensional; existence of B with $B^2 = I$, $\text{Tr}(B) = 0$, $AB + BA = 0$
- ☐ Existence of 3-dimensional invariant subspace for any operator on 4-dim $\mathbb{C}$ space
- **Matrices over Finite Fields**
 - ☐ $GL_n(\mathbb{F}_q)$: number of invertible matrices $= (q^n - 1)(q^n - q) \cdots (q^n - q^{n-1})$
 - ☐ Number of 2×2 matrices over $\mathbb{F}_5$ of rank 1: subtract invertible + zero from total 5^4
 - ☐ Subspaces of $\mathbb{F}_q^{2n}$ containing a given n -dim subspace (Grassmannian counting)
- **Integer Matrices and Minimal Polynomials**
 - ☐ Non-zero 3×3 integer matrix: char poly over $\mathbb{Q}$; if $f(\lambda_i) = 0$ for all i and f cubic $\Rightarrow f(A) = 0$ (Cayley-Hamilton suffices)
 - ☐ Does $f \in \mathbb{Q}[X]$ quadratic exist with all eigenvalues as roots? Only if poly factors
 - ☐ Minimal polynomial divides any annihilating polynomial

3

Statistics & Probability 30 Questions across both papers | Part B: 5 | Part C: 8 | **Tier 1 — Highest Priority**

Sub-topics

• Point Estimation

- ☐ Maximum likelihood estimation (MLE): discrete and continuous families
- ☐ Method of moments (MOM): equate population moments to sample moments
- ☐ Sufficient statistics: Factorisation theorem (Neyman-Fisher)
- ☐ Minimal sufficient statistics; completeness
- ☐ Ancillary statistics: definition; examples (ratios, differences of normals)
- ☐ Unbiased estimators; UMVUE via Lehmann-Scheffé; Rao-Blackwellisation
- ☐ Cramér-Rao lower bound; Fisher information
- ☐ Consistency: $\hat{\theta}_n \xrightarrow{P} \theta$; MLE under regularity is consistent

• Hypothesis Testing

- ☐ Neyman-Pearson lemma; most powerful tests; UMP tests
- ☐ Likelihood ratio test (LRT); Wilks' theorem ($-2 \log \Lambda \xrightarrow{d} \chi^2$)
- ☐ Type I / Type II errors; significance level; p -value
- ☐ One-sided and two-sided tests; uniformly most powerful (UMP) tests
- ☐ Sign test: K = number of obs $> \theta_0$; $K \sim \text{Bin}(n, 1/2)$ under H_0
- ☐ Non-parametric tests: Wilcoxon, Mann-Whitney (basics)
- ☐ Test statistic F for regression significance; relation to R^2
- ☐ Confidence intervals from pivot quantities; relation to hypothesis tests

• Bayesian Estimation

- ☐ Prior, likelihood, posterior: Bayes' theorem in density form
- ☐ Beta-Binomial conjugate: prior $\text{Beta}(\alpha, \beta) + \text{Binomial}(n, \theta) \Rightarrow$ posterior $\text{Beta}(\alpha + x, \beta + n - x)$
- ☐ Posterior mean $= \frac{x + \alpha}{n + \alpha + \beta}$; posterior mean $\geq$ prior mean $\Leftrightarrow x/n \geq \alpha/(\alpha + \beta)$
- ☐ Improper prior $\pi(\theta) \propto 1$: posterior is proportional to likelihood
- ☐ Normal-Normal conjugate: $X|\theta \sim N(\theta, \sigma^2)$, prior $\theta \sim N(\mu_0, \tau^2)$; posterior mean is weighted average
- ☐ Posterior mode vs. MLE; posterior variance vs. $\text{Var}(X|\theta)$

• Probability Distributions

- ☐ Exponential, Gamma, Beta, Normal, Binomial, Poisson, Uniform, Cauchy
- ☐ MGF identification: $M_X(t) = e^{3(e^t - 1)} \Rightarrow X \sim \text{Poisson}(3)$; $M_Y(t) = (\frac{1}{2}e^{-3t} + \frac{1}{2}e^{3t})^2$ etc.
- ☐ CDF, survival function $S(t)$, hazard rate $\lambda(t) = f(t)/S(t)$, cumulative hazard $H(t)$
- ☐ Series system lifetime: $Y = \min\{T_1, \dots, T_n\}$; $P(Y > t) = (S(t))^n$
- ☐ Reliability / survival analysis: $\lambda(t) = 1/(1+t) \Rightarrow H(t) = \ln(1+t)$, $S(t) = 1/(1+t)$
- ☐ Mixture distributions (brand I exponential + brand II gamma)
- ☐ Discrete distribution with PMF $P(X = k) = \theta^k/k!$ type; MLE from observed counts

• Convergence and Limit Theorems

- ☐ Convergence in distribution, probability, almost sure, L^p ; hierarchy
- ☐ Weak Law of Large Numbers; Strong LLN (Kolmogorov)
- ☐ Central Limit Theorem; CLT for i.i.d. with finite variance
- ☐ Extreme value theory: $T_n = \max\{X_1, \dots, X_n\} - \ln n \xrightarrow{d}$ Gumbel distribution
- ☐ Gumbel CDF: $F(x) = e^{-e^{-x}}$; $P(Y \leq 0) = e^{-1}$; median $= \ln(\ln 2)$
- ☐ Remainder $X_n = Y_n \bmod m$: as $n \rightarrow \infty$, converges to Uniform on $\{0, 1, \dots, m-1\}$

- **Multivariate Distributions**

- ☐ Multivariate normal $N_p(\mu, \Sigma)$: marginals, conditionals, affine transformations
- ☐ Sample covariance S : distribution is Wishart $W_p(n-1, \Sigma)$
- ☐ $E(S^{-1}) = \frac{n-1}{n-p-2} \Sigma^{-1}$ (for $n > p+2$); specific: $n=20, p=12$ gives $E(S^{-1}) = \frac{19}{6} \Sigma^{-1}$
- ☐ $\text{tr}(19\Sigma^{-1}S) \sim \chi^2_{12 \times 19} = \chi^2_{228}$
- ☐ Multiple correlation coefficient between Y_1 and (Y_2, Y_3)
- ☐ Partial correlation $\rho_{YZ.X}$: formula via inverse of covariance matrix
- ☐ $19(\bar{X}^\top S \bar{X})(\bar{X}^\top \Sigma \bar{X})^{-1} \sim \chi^2_{19}$

- **Markov Chains**

- ☐ Definitions: homogeneous MC, transition matrix P , Chapman-Kolmogorov
- ☐ Communication classes; irreducibility; period of a state
- ☐ Recurrence vs. transience; positive recurrence; null recurrence
- ☐ Stationary distribution π : $\pi P = \pi$, $\sum \pi_i = 1$; existence and uniqueness for irreducible positive-recurrent chains
- ☐ Mean recurrence time: $m_i = 1/\pi_i$
- ☐ Absorbing states; hitting probabilities; expected hitting times
- ☐ Specific 4×4 TPM: identify recurrent/transient states; stationary distribution; ergodic theorem
- ☐ Infinitely many stationary distributions when multiple closed communicating classes exist

- **Linear Regression and ANOVA**

- ☐ Simple linear regression $Y_i = \beta x_i + \varepsilon_i$: OLS estimator; unbiased; variance
- ☐ Minimise $E(\sum a_i Y_i - \beta)^2$: optimal $a_i^* = x_i / (\sum x_j^2)$; this has smaller variance than OLS
- ☐ Multiple regression $Y = X\beta + \varepsilon$; BLUE (Gauss-Markov); $\hat{\beta} = (X^\top X)^{-1} X^\top Y$
- ☐ R^2 , Adjusted R^2 , F -statistic; relation: $F = \frac{R^2/p}{(1-R^2)/(n-p-1)}$
- ☐ Decomposition $Y^\top AY = Y^\top BY + Y^\top CY$ (total=regression+residual); $Y^\top BY, Y^\top CY$ always independent
- ☐ Design of experiments: 2^k factorial, confounding, block design

- **Sampling Theory**

- ☐ SRSWR vs. SRSWOR: $\text{Var}(\bar{y}_{WR}) = \frac{S^2}{n}$, $\text{Var}(\bar{y}_{WOR}) = \frac{S^2}{n}(1 - \frac{n}{N})$; ratio = $1 - f$
- ☐ Stratified random sampling: $\bar{y}_{st} = \sum (N_h/N) \bar{y}_h$; unbiased estimator formula
- ☐ $N=100$, SRSWR vs. SRSWOR: $\text{Var}(T_1) = 9\text{Var}(T_2) \Rightarrow n=89$

- **Queuing Theory**

- ☐ $M/M/1$, $M/M/c$ systems; birth-death process; balance equations
- ☐ $M/M/2$ with capacity 3, $\lambda=4$, $\mu=1$: compute $p_0=4/29$, $p_3=12/29$; $p_0+p_3=16/29$ (verify from answer key)
- ☐ Long-run fraction of time system is empty/full; throughput; mean queue length

4

Complex Analysis 26 Questions across both papers | Part B: 4 | Part C: 5-6 |
Tier 2

Sub-topics

- **Holomorphic and Analytic Functions**
 - ☐ Cauchy-Riemann equations; harmonic functions; harmonic conjugate
 - ☐ Power series; radius of convergence; analytic $\Leftrightarrow$ holomorphic
 - ☐ Analytic continuation; identity theorem (zeros with limit point $\Rightarrow f \equiv 0$)
 - ☐ Function $f : \mathbb{C} \rightarrow \mathbb{C}$ holomorphic with $f(2z) = f(z)/(1 - f(z)^2)$: only one such function
- **Cauchy's Theorem and Applications**
 - ☐ Cauchy's integral theorem (simply/multiply connected domains)
 - ☐ Cauchy's integral formula; derivatives via contour integrals
 - ☐ Morera's theorem; Goursat's theorem
 - ☐ Contour integrals on $\gamma_R(t) = Re^{2\pi it}$: $\int_{\gamma_R} \tan z \, dz$, $\int_{\gamma_R} f(z) \, dz = 0$ conditions
 - ☐ $\int_{\gamma_1} \frac{dz}{z \sin z} = 0$ (no singularity inside unit circle); $\int_{\gamma_2} \frac{dz}{z \sin z}$ when $\gamma_2 = 1 + i + e^{it}$
- **Singularities and Laurent Series**
 - ☐ Removable singularities (Riemann), poles (order), essential singularities (Picard)
 - ☐ Laurent series expansion; principal part; residue computation
 - ☐ Casorati-Weierstrass: near essential singularity, image is dense
 - ☐ Function $g(z) = f(z) - 1/z$ (f entire): pole at 0; no zeros on $|z| = 1$; finitely many zeros; $\max_{|z|=1} |g| \geq 1$
- **Residue Theorem**
 - ☐ Residue theorem: $\oint f \, dz = 2\pi i \sum \text{Res}$
 - ☐ Computing residues: simple poles ($\lim_{z \rightarrow a} (z-a)f(z)$), higher order ($\lim_{z \rightarrow a} \frac{d^{n-1}}{dz^{n-1}} [(z-a)^n f(z)] / (n-1)!$)
 - ☐ $\int_{\gamma_2} \tan z \, dz = -4\pi i$ (poles of $\tan z$ at $\pm\pi/2$; both inside γ_2 of radius 2 centred at $2+i$)
 - ☐ Application to real integrals: $\int_{-\infty}^{\infty} R(x)dx$, $\int_0^{\infty}$ type
- **Entire Functions**
 - ☐ Liouville's theorem; entire bounded $\Rightarrow$ constant
 - ☐ Picard's theorem: entire non-constant misses at most one value
 - ☐ Image constraints: $f(\mathbb{C}) \subset \{x + iy : y = x + 1\} \Rightarrow f(z)/z \rightarrow 0$ (not f constant)
 - ☐ Entire f with prescribed derivatives: $f^{(n)}(0) = n!/n^n \Rightarrow$ exists (convergent power series); $f^{(n)}(0) = n!n^n \Rightarrow$ diverges
 - ☐ Holomorphic sequence $f_n(z) = \sum_{k=1}^n h(z^k)$ on $\{|z| > 1\}$: converges pointwise
- **Möbius Transformations**
 - ☐ $f(z) = (az + b)/(cz + d)$; cross-ratio; maps circles/lines to circles/lines
 - ☐ Image of $[0, 1]$ under $f(z) = (z - i)/(z + i)$: arc $\{e^{i\theta} : \theta \in [\pi, 3\pi/2]\}$
 - ☐ Three-point determination; automorphisms of $\mathbb{D}$, $\mathbb{H}$, $\mathbb{C}$
- **Schwarz Lemma and Conformal Maps**
 - ☐ Schwarz-Pick lemma: $f : \mathbb{D} \rightarrow \mathbb{D}$ hol., $|f'(a)| \leq (1 - |f(a)|^2)/(1 - |a|^2)$
 - ☐ $f(-1/2) = 0$: $|f(-1/5)| \leq 1/3$ (Schwarz-Pick estimate); $|f'(-1/2)| \leq 4/3$
 - ☐ Riemann mapping theorem (statement); automorphisms of $\mathbb{D}$

5

Abstract Algebra 24 Questions across both papers | Part B: 4 | Part C: 5 | **Tier 2**

Sub-topics

• Group Theory — Basics

- ☐ Group axioms; order of element; cyclic groups $\mathbb{Z}_n$; generators
- ☐ Subgroups; cosets; Lagrange's theorem
- ☐ Normal subgroups; quotient groups; group homomorphism theorems
- ☐ Conjugacy classes; centre $Z(G)$; class equation
- ☐ Groups of order 8: $\mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$, $\mathbb{Z}_2^3$, D_4 , Q_8 ; elements of order 4 in each
- ☐ Number of subgroups $S(G)$; if $S(G) = 5 \Rightarrow G$ cyclic; if $S(G) = 4 \Rightarrow |G| = p^m$ necessary?

• Sylow Theorems and Structure

- ☐ Sylow p -subgroups; Sylow theorems (existence, conjugacy, count)
- ☐ p -groups; centre is non-trivial; p -groups of order p^2 are abelian
- ☐ Non-abelian groups: $f(a, b) = ab$ is **not** a group homomorphism; $f : G \times G \rightarrow G$
- ☐ If every divisor of $|G|$ has a subgroup $\Rightarrow$ at least 3 normal subgroups
- ☐ $H_G = \{g : g^{15} = e\}$: $|H_G|$ can be $4n + 1$ for $n \geq 3$; structure

• Permutation Groups

- ☐ S_n , A_n ; cycle type; sign of permutation; transpositions
- ☐ Orbits, stabilisers, Burnside (Cauchy-Frobenius) lemma
- ☐ S_{17} : for subgroup G and $x \in X$, stabiliser $S_G(x)$; Burnside's lemma $\Rightarrow$ integer orbit count; *not* necessarily isomorphic for different x
- ☐ Homeomorphisms of finite topological space $S = \{1, 2, 3, 4, 5\}$ with $\tau = \{\emptyset, \{1\}, S\}$: 24 (fix 1, permute 2–5 freely)

• Ring Theory

- ☐ Integral domains, fields; zero divisors; characteristic
- ☐ Ideals: principal, prime, maximal; quotient rings
- ☐ PIDs, UFDs, Euclidean domains; their hierarchy
- ☐ Valuation rings: $A \subset \mathbb{Q}$ with $r \in A$ or $1/r \in A$ for all nonzero r ; unique maximal ideal; every finitely-generated ideal is principal; **not** necessarily Euclidean
- ☐ $2\mathbb{Z}[i]$ maximal in $\mathbb{Z}[i]$? YES (norm-2 Gaussian prime)
- ☐ $X\mathbb{C}[X, Y]$ maximal in $\mathbb{C}[X, Y]$? NO (not maximal; $\mathbb{C}[X, Y]/X\mathbb{C}[X, Y] \cong \mathbb{C}[Y]$ not a field)
- ☐ $(\sqrt{2} - 1)\mathbb{Z}[\sqrt{2}]$ maximal in $\mathbb{Z}[\sqrt{2}]$? YES (norm-1 unit $\Rightarrow$ ideal = whole ring? Check)

• Polynomial Rings and Irreducibility

- ☐ Eisenstein irreducibility criterion; reduction mod p criterion
- ☐ $x^2 + xy^2 - x - y + 1$: irreducible in $\mathbb{Q}[x, y]$
- ☐ $x^3 + 100x^2 + 25x + 10$: irreducible in $\mathbb{Z}[x]$ (Eisenstein mod 5)
- ☐ $2x^2 + 3xy + y^2 + 3x + 2y + 1 = (2x + y + 1)(x + y + 1)$: *reducible* in $\mathbb{Q}[x, y]$
- ☐ $x^4 + 4x^2 + 3 = (x^2 + 1)(x^2 + 3)$: reducible in $\mathbb{Z}[x]$
- ☐ Gauss's lemma; primitive polynomials; $\mathbb{Z}[x]$ is a UFD

• Field Extensions and Galois Theory

- ☐ Algebraic / transcendental extensions; degree $[L : K]$; tower law
- ☐ Splitting fields; normal and separable extensions; Galois closure
- ☐ Galois group $\text{Gal}(K/\mathbb{Q})$; fundamental theorem of Galois theory
- ☐ Irreducible $f \in \mathbb{Q}[x]$ of prime degree p with exactly 2 non-real roots $\Rightarrow G = S_p$: proof sketch (transposition from complex conjugation + p -cycle from irreducibility / prime degree)
- ☐ Finite fields $\mathbb{F}_q$: existence, uniqueness; Frobenius automorphism

- **Number Theory (within Algebra)**

- ☐ Wilson's theorem: p prime $\Rightarrow p \mid (p-1)! + 1$; so $p \mid 1 + (p-1)!$; *not* $p \nmid 1 + (p-1)!$
- ☐ Fermat's little theorem: $p \mid a^p - a$ for all a, p ; $(1234)^{p-1} \equiv 1 \pmod{p}$ for $\gcd(1234, p) = 1$; true for primes $p > 700$ (none divide 1234)
- ☐ $p \nmid a^p - a$ is FALSE (Fermat's little theorem); always $p \mid a^p - a$
- ☐ Existence of $a \in \mathbb{Z}$, prime $p > 11$ with $p \nmid a^p - a$: FALSE (Fermat)
- ☐ $(p^2)!/(p!)^2$: divisibility by prime p (involves Kummer's theorem / p -adic valuations)

6

Ordinary Differential Equations 22 Questions across both papers | Part B: 3 | Part C: 4 | Tier 2

Sub-topics

• IVPs: Existence and Uniqueness

- ☐ Picard-Lindelöf theorem: f Lipschitz in $y \Rightarrow$ unique local solution
- ☐ Non-uniqueness: $y' = y^{1/3}$, $y(0) = 0$ has infinitely many solutions
- ☐ $y' = f(y)$, $f(y) = y \cos(1/y)$ for $y \neq 0$, $f(0) = 0$: unique solution $y \equiv 0$ (P) is TRUE
- ☐ IVPs with $y \cos(1/y)$: Lipschitz fails at 0 but still unique; check via phase-plane
- ☐ Taylor series solution: $r(x)$ real-analytic near $x = 0 \Rightarrow$ solution has Taylor series; r flat (e^{-1/x^2}) $\Rightarrow$ solution is flat, has Taylor series $\equiv 0$

• Linear ODEs — Second Order

- ☐ General solution = particular + homogeneous; variation of parameters
- ☐ Wronskian: $W(y_1, y_2)(x) = W_0 \exp(-\int p(x)dx)$; constant if $p = 0$
- ☐ One known solution $\varphi(x) = x$; reduction of order; find second solution
- ☐ ODE $y'' - (2/x^2 + 1/x)(xy' - y) = 0$, $\varphi = x$ known $\Rightarrow$ general sol. $(a + be^{-2x})x$
- ☐ Wronskian at specific point: $W(y_1, y_2)(1/2) = 4$, find $y'_2(0)$

• Sturm-Liouville Problems

- ☐ $(r(x)y')' + (\lambda w(x) - q(x))y = 0$; eigenvalues $\lambda_n \nearrow \infty$; eigenfunctions orthogonal w.r.t. weight w
- ☐ $\frac{d}{dx}(x \frac{dy}{dx}) + \frac{\lambda}{x}y = 0$, $1 < x < e^{2\pi}$, $y(1) = y(e^{2\pi}) = 0$: eigenvalues $\lambda_n = n^2/4\pi^2$; $\sum 1/\lambda_n = \pi^2/12$ (Basel-type)
- ☐ Expansion in eigenfunctions; Parseval's identity applied to series $\sum 1/\lambda_n$
- ☐ Sturm comparison theorem: zeros of solutions interleave
- ☐ ODE $y'' + r(x)y = 0$ with $r(x) = x^3 \sin(1/x)$: Wronskian is constant; solutions have infinitely many zeros approaching $\pm\infty$ (not 0); in any bounded interval only finitely many zeros

• Frobenius Method / Power Series

- ☐ Regular singular points; indicial equation; two independent Frobenius solutions
- ☐ $x^2 y'' + \frac{(1+\sin x)x}{2} y' - \frac{3}{2}(\cos x)y = 0$: indicial roots at $x = 0$ are $r_1 = 3/2, r_2 = -1/2$ (differs by integer? check logs); $y_2(0) = 0 \Rightarrow y_2 \sim x^{-1/2}$ -type; various limit ratios computed
- ☐ When roots differ by integer: possible log terms; careful case analysis

• Integral Equations

- ☐ Volterra IE of 2nd kind: $y(x) = f(x) + \lambda \int_0^x K(x, t)y(t)dt$; successive approximation
- ☐ Fredholm IE: $y(x) = f(x) + \lambda \int_a^b K(x, t)y(t)dt$; eigenvalues of kernel; Fredholm alternative
- ☐ Degenerate (separable) kernels: $K(x, t) = \sum_i \phi_i(x)\psi_i(t)$; reduce to linear algebra
- ☐ $y(x) = e^x + e + 1 + \frac{1}{3} \int_0^1 y(t)dt$: constant kernel, resolvent $R(x, t; \lambda) = \lambda/(1 - \lambda)$; $y(1) = 3e + 1$
- ☐ $y(x) = 3 + \int_0^x (x - t)u(t)dt$: convert to ODE $u'' - u = 0$; solve with $u(0) = 3$, $u'(0) = 0 \Rightarrow u = 3 \cosh x$; then $u(\pi) + u'(\pi) = 3e^\pi$
- ☐ IE $y(x) = x^2 + 2 \int_0^1 xty(t)dt$: degenerate kernel $K = xt$; $y(x) = x^2 + cx$ with $c = 2 \int_0^1 ty(t)dt$; solve for c ; verify $y(-1) + y(1) = 1$

• Systems and Stability (basics)

- ☐ Autonomous systems; phase portraits; equilibria
- ☐ Linearisation; stable/unstable/saddle equilibria
- ☐ Lagrangian mechanics ODE: Euler-Lagrange $\Rightarrow$ equations of motion; conservation laws

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Partial Differential Equations 18 Questions across both papers | Part B: 2–3 | Part C: 4 | Tier 3

Sub-topics

- **First-Order PDEs**
 - ☐ Method of characteristics: $u_t + u_x = 0$, $u_t + a(x, t)u_x = b$; Cauchy data
 - ☐ $(y + u)u_x + yu_y = x - y$, $u(x, 1) = 1 + x$: characteristics; $u(1, 1) = 2$ (verify by characteristics)
 - ☐ Transport equation IBVP: $u_t + u_x = 0$, $u(x, 0) = e^x$, $u(0, t) = 1 - \sin t$: unique sol for $t < x$; for $t > x$ compatibility condition
 - ☐ Quasi-linear equations; genuine nonlinearity; shocks and rarefaction waves
- **Wave Equation**
 - ☐ d'Alembert's formula: $u(x, t) = \frac{1}{2}[f(x+t) + f(x-t)] + \frac{1}{2} \int_{x-t}^{x+t} g(s)ds$
 - ☐ $u_{tt} - u_{xx} = 0$, $u(x, 0) = 1 + x^2$, $u_t(x, 0) = x + 1$: $u(1, 1) = 5$
 - ☐ Non-homogeneous wave: $u_{tt} - u_{xx} = e^{-t}$; Duhamel's principle; $u(x_0, t_0) \geq 4$ for large t_0
 - ☐ IBVP energy: $E(t) = \int (u_t^2 + u_x^2)dx$; $\dot{E} = -2 \int u_t^2 dx \leq 0$; hence $E(t) \leq E(0) = \pi$
- **Heat Equation**
 - ☐ $u_t = ku_{xx}$; separation of variables; Fourier series solution
 - ☐ $u_t = 16u_{xx} + 2$, $u_x(0, t) = 0$, $u(7, t) = 0$: steady state $u^* = \frac{1}{16}(49 - x^2)$; $u(x, t) \rightarrow u^*$ as $t \rightarrow \infty$
 - ☐ Maximum principle: $\max u$ attained on boundary (parabolic region)
 - ☐ Fundamental solution (heat kernel); Green's function approach
- **Laplace and Elliptic Equations**
 - ☐ Laplace equation $\Delta u = 0$; harmonic functions; maximum principle
 - ☐ Dirichlet BVP on disk: $u|_{\partial\Omega} = e^{x+y} \Rightarrow$ unique solution; $\min_{|z| \leq 1} e^{x+y} = e^{-\sqrt{2}} < \frac{1+e}{2} < e^{\sqrt{2}} = \max$ so $u = \frac{1+e}{2}$ at some interior pt
 - ☐ Poisson's formula for disk; mean value property
 - ☐ Unique solvability of Dirichlet problem on bounded domain; Green's function
- **Energy Methods**
 - ☐ IBVP $u_{tt} - u_{xx} + u_t = 0$: $\dot{E} = -2 \int u_t^2 dx \leq 0$; $E(t) \leq E(0) = \pi$, and $E(t) \leq \pi/2$ since E strictly decreasing + $E \rightarrow 0$
 - ☐ Uniqueness via energy: assume two solutions u, v ; $w = u - v$ satisfies zero data; $E_w(t)$ is non-increasing, starts at 0 $\Rightarrow w \equiv 0$

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Topology 16 Questions across both papers | Part B: 2 | Part C: 4 | Tier 3

Sub-topics

- **Topological Spaces and Continuity**
 - ☐ Open/closed sets; basis; sub-basis; interior, closure, boundary, limit points
 - ☐ Continuous maps; homeomorphisms; topological properties (preserved under homeomorphism)
 - ☐ Finite complement topology on $\mathbb{R}$: not Hausdorff; not regular; not normal
 - ☐ Coarser/finer topologies; identity map from $(\mathbb{R}, \tau_E) \rightarrow (\mathbb{R}, \tau)$ continuous $\Rightarrow \tau \subset \tau_E \Rightarrow$

- $(\mathbb{R}, \tau)$ is regular? *Yes* (inherits separation from $\mathbb{R}$)
- ☐ Identity from $(\mathbb{R}, \tau) \rightarrow (\mathbb{R}, \tau_E)$ continuous $\Rightarrow \tau_E \subset \tau \Rightarrow (\mathbb{R}, \tau)$ is Hausdorff
 - ☐ Homeomorphisms of $S = \{1, 2, 3, 4, 5\}$ with $\tau = \{\emptyset, \{1\}, S\}$: must fix 1; permute $\{2, 3, 4, 5\}$ freely $\Rightarrow 4! = 24$
 - **Compactness**
 - ☐ Compact: every open cover has finite subcover; sequential compactness equivalent in metric spaces
 - ☐ Tychonoff's theorem: arbitrary product of compact spaces is compact
 - ☐ $X = \prod_{c>0} [0, c]$ with product topology: compact (Tychonoff); Hausdorff; connected; **not** metrizable (uncountable product)
 - ☐ Continuous image of compact is compact; closed subspace of compact is compact
 - ☐ Compact Hausdorff $\Rightarrow$ normal; compact metric $\Rightarrow$ complete + totally bounded
 - **Connectedness and Path-Connectedness**
 - ☐ Product of connected spaces is connected (including uncountable products)
 - ☐ Path-connected $\Rightarrow$ connected; converse fails (topologist's sine curve)
 - ☐ Continuous image of connected is connected
 - ☐ $[0, c]$ is connected for all $c > 0$; arbitrary product is connected
 - **Separation Axioms**
 - ☐ T_0, T_1, T_2 (Hausdorff), T_3 (regular), T_4 (normal): definitions and implications
 - ☐ $(\mathbb{R}, \tau_E)$: T_4 (normal); metric spaces are normal
 - ☐ Urysohn's lemma; Tietze extension theorem (for normal spaces)
 - ☐ Product topology and separation: product of Hausdorff is Hausdorff; product of normal need not be normal
 - ☐ Compact Hausdorff space: statement S (neighbourhood of $\{x\} \times Y$ when Y compact) is TRUE $\Leftrightarrow Y$ compact
 - **Metrizability**
 - ☐ Second countable + regular $\Rightarrow$ metrizable (Urysohn)
 - ☐ Uncountable product of non-trivial spaces: not second countable $\Rightarrow$ not metrizable
 - ☐ $\prod_{c>0} [0, c]$: uncountable index set $\Rightarrow$ not metrizable

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Functional Analysis 12 Questions across both papers | Part B: 1 | Part C: 3 | Tier 3

Sub-topics

• Hilbert Spaces

- ☐ Inner product, norm, Cauchy-Schwarz; completeness $\Rightarrow$ Hilbert space
- ☐ Orthonormal systems; Bessel's inequality; Parseval's identity; ONB
- ☐ Best approximation from closed subspace; projection theorem
- ☐ $\ell^2 = \{(a_n) : \sum |a_n|^2 < \infty\}$ with $\langle a, b \rangle = \sum a_n b_n$
- ☐ Set $H = \{(a_n) \in \ell^2 : |a_n| \leq 1/n \forall n\}$: bounded (by $\|a\|^2 \leq \sum 1/n^2 = \pi^2/6$); convex; **not** a subspace (not closed under scalar mult.); contains **no** ONB (standard basis $e_n \notin H$)

• Banach Spaces and Operators

- ☐ Normed spaces, Banach spaces; L^p spaces; $C[0, 1]$ with supremum norm
- ☐ Bounded linear operators; operator norm; dual space
- ☐ Open mapping theorem; closed graph theorem; uniform boundedness (Banach-Steinhaus)
- ☐ Compact operators on Hilbert spaces; spectral theory basics

• Compact Sets in $C[0, 1]$

- ☐ Arzelà-Ascoli theorem: equicontinuous + uniformly bounded $\Rightarrow$ precompact in $C[0, 1]$
- ☐ $f_n(x) = n(1 - nx)$ for $0 \leq x \leq 1/n$, 0 otherwise: uniformly bounded (by 1); converges pointwise to 0; converges uniformly to 0; $\{f_n\}$ is compact in $(C[0, 1], \|\cdot\|_\infty)$ (equicontinuous? Check Arzelà-Ascoli carefully)
- ☐ $f_n = \sin(nx)/n$: uniform convergence to 0, compact

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Calculus of Variations & Classical Mechanics 11 Questions across both papers | Part B: 2 | Part C: 3 | Tier 3

Sub-topics

• Euler-Lagrange Equation

- ☐ $J(y) = \int_a^b F(x, y, y') dx$; E-L: $F_y - \frac{d}{dx} F_{y'} = 0$
- ☐ $J(y) = \int_0^4 (y')^2 / y^2 dx$, $y(0) = 1$, $y(4) = e^8$: extremal $y = e^{2x}$; $y(\ln 3) = 9$
- ☐ $J[y] = \int_0^{\pi/4} ((y')^2 - 4y^2 + 2xy) dx$, $y(0) = 0$, $y(\pi/4) = 1$: extremal involves $\sin 2x$ and x terms
- ☐ $J[y] = \int_0^1 ((y')^2 \sin x + 2y \cos x) dx$: E-L gives $(2y' \sin x)' = 2 \cos x$; solve with BCs; $y'(0) = 1$, $y'(1/2) = 1$
- ☐ Transversality conditions for free endpoints
- ☐ Natural boundary conditions; higher-order functionals

• Isoperimetric Problems

- ☐ Lagrange multiplier method for constraints $\int G(x, y, y') dx = C$
- ☐ $J(y) = \int_0^1 x^2 (y')^2 dx$, $y(0) = 0$, $y(1) = 1$, $\int_0^1 y^2 dx = 1/7$: extremal $y = x^3$; $y'(1/2) = 3/4$, $y'(1/\sqrt{3}) = 1$, $y'(1/3) = 1/3$
- ☐ Dido's problem; isoperimetric inequality

• Classical Mechanics

- ☐ Lagrangian $L = T - V$; Euler-Lagrange equations of motion

- ☐ $L = (\dot{q}_1)^2 + (\dot{q}_2)^2 + (q_1)^2 + \dot{q}_1\dot{q}_2$: momenta $p_1 = 2\dot{q}_1 + \dot{q}_2$, $p_2 = 2\dot{q}_2 + \dot{q}_1$; equations of motion
- ☐ Hamiltonian $H = \sum p_i\dot{q}_i - L$; Hamilton's equations
- ☐ Canonical transformations; generating functions $F(q, P)$, $G(p, Q)$
- ☐ $F(q, P) = q^2e^P$: $p = \partial F/\partial q = 2qe^P$, $Q = \partial F/\partial P = q^2e^P$; hence $p = 2\sqrt{Q}e^P$; $G(p, Q) = -Q[1 + \ln(p^2/(4Q))]$

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Numerical Methods 8 Questions across both papers | Part B: 2 | Part C: 2 | Tier 3

Sub-topics

• Root-Finding Methods

- ☐ Newton-Raphson: $x_{n+1} = x_n - f(x_n)/f'(x_n)$; quadratic convergence ($p = 2$) at simple zeros
- ☐ Order p and asymptotic error constant: $|e_{n+1}|/|e_n|^p \rightarrow C$
- ☐ $\sin x - 1 = 0$, root at $\pi/2$: $f(\pi/2) = 0$, $f'(\pi/2) = 0$, $f''(\pi/2) = -1 \Rightarrow$ double zero; N-R at double zero converges linearly ($p = 1$); $\lim |e_{n+1}|/|e_n| = 1/2$
- ☐ Bisection, secant, regula falsi; convergence orders

• Iterative Methods for Linear Systems

- ☐ Jacobi and Gauss-Seidel methods; splitting $A = D + L + U$
- ☐ Convergence condition: spectral radius $\rho(T) < 1$ where $T = -(D + L)^{-1}U$ (GS) or $T = -D^{-1}(L + U)$ (Jacobi)
- ☐ $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$: GS converges ($\rho < 1$); $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$: GS diverges; $\begin{pmatrix} 1 & 5 \\ 1 & 10 \end{pmatrix}$: GS converges; $\begin{pmatrix} -5 & 2 \\ 1 & -4 \end{pmatrix}$: GS converges
- ☐ Strictly diagonally dominant matrices $\Rightarrow$ both Jacobi and GS converge

• Interpolation

- ☐ Lagrange interpolating polynomial; Newton's divided difference formula
- ☐ Degree of interpolating polynomial: $n + 1$ points $\Rightarrow$ degree $\leq n$
- ☐ Interpolation at $x \in \{-2, -1, 0, 1, 2\}$, $y \in \{-1, 3, 1, -1, 3\}$: $p(1/2) = -3/8$ (compute via Newton's table)
- ☐ Error bound: $|f(x) - p_n(x)| \leq \frac{\max |f^{(n+1)}|}{(n+1)!} \prod |x - x_i|$

• Splines

- ☐ Cubic splines: piecewise cubic, C^2 continuity at knots
- ☐ $s(x)$ piecewise on $[0, 1]$, $[1, 3]$, $[3, 4]$: match s, s', s'' at $x = 1$ and $x = 3$; solve for a, b, c ; $2a + b + 2c = 4$ (from both papers)
- ☐ Natural spline: $s'' = 0$ at endpoints; clamped spline: s' prescribed at endpoints

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Linear Programming & Operations Research 3 Questions across both papers | Part B: 1 | Part C: 1 | Tier 4

Sub-topics

• Linear Programming

- ☐ Standard form; feasible region (convex polytope); vertices/extreme points
- ☐ Simplex method: basic feasible solutions; pivot operations; optimality condition
- ☐ Graphical method for 2-variable LP: find corner points; evaluate objective
- ☐ LP: $\max z = x + y$ s.t. $5x + 3y \leq 30$, $2x + 6y \leq 25$, $2x - y \leq 8$, $x, y \geq 0$: optimal $z = 74/11$ at vertex $(15/11, \dots)$
- ☐ LP: $\max 3x_1 + 4x_2$ s.t. $3x_1 + 2x_2 \leq 12$, $3x_1 + 5x_2 \leq 15$, $2x_1 - x_2 \geq 0$, $x_2 \leq 2$: optimal value = 14

-
- ☐ Duality: dual problem; strong duality theorem; complementary slackness
 - ☐ Unbounded feasible region; infeasible LP; unbounded objective

A

PART A — General Aptitude 20 Questions | Attempt 15 | +2 / -0.5

Sub-topics

- **Mathematical Reasoning & Arithmetic**
 - ☐ Ratio and proportion; mixture problems (alloy ratios)
 - ☐ Percentages; percentage change; profit and loss
 - ☐ Averages: arithmetic mean, geometric mean (*GM* of scaled data)
 - ☐ Time, speed, distance; RPM and circular motion
 - ☐ Number theory: LCM, HCF, divisibility; digits and unit's place computations
 - ☐ Sequences satisfying recurrence: $a_{i-1} + a_i + a_{i+1} = C$; find a_{106}
 - ☐ Counting: 5-digit numbers divisible by 5 from given digit set
 - ☐ Stock market: two-day loss/recovery; fraction remaining of initial investment
- **Logical Reasoning**
 - ☐ Syllogisms: All A are B, All B are C $\Rightarrow$ All A are C; valid conclusions
 - ☐ Blood relations: "that man's father is my father's son" type
 - ☐ Seating / ordering: Jiten, Anwar between Ramesh and Prakash but senior to Sam
 - ☐ Coding-decoding: DELTOID $\rightarrow$ 3152893; LOTION $\rightarrow$?
 - ☐ Venn diagrams: three-sport players; percentage playing exactly two sports
 - ☐ Colour matching: three friends, three shirt colours, no colour starts with person's initial
 - ☐ Two-digit number ab : $ab - (a + b)$ always divisible by 9; result always $= 9a - 9$ or $9b$?
 - ☐ Statements: All Booklets are Manuals, All Manuals are Catalogues $\Rightarrow$ All Booklets are Catalogues
- **Data Interpretation**
 - ☐ Pie chart + bar graph: market share % and production (thousands) for 5 companies; profit $\propto$ market/production ratio
 - ☐ Line graphs: tree count in two different years; maximum relative change
 - ☐ Venn diagram: overlapping circles with numbers; total players, exactly-two-sport players
- **Geometry & Measurement**
 - ☐ Minimum mirror height: half the person's height (independent of distance from mirror)
 - ☐ Volume: sphere immersed in cylinder; water level rise $\Delta h = \frac{4r^3}{3R^2}$
 - ☐ Semi-circles rearranged: three identical semi-circles with overlap 10; diameter = 20
 - ☐ Circle quartered and rejoined: area remains $= \pi r^2 = \pi$ (since $r = 1$); area of shape = 1 (area of square from arcs)
 - ☐ Distance: Kavita walks 200 N, 45 right 70 m, 90 right 70 m; shortest distance ≈ 223 m
 - ☐ Jogging on circular track: Rahul full circle, father half then straight back; ratio $2\pi/(\pi+2)$
- **General Science Reasoning**
 - ☐ Cooling curves: two bars at different temperatures cooling in same surroundings
 - ☐ Density-mass table: largest volume from $V = m/\rho$
 - ☐ Temperature gradient: path along which temperature change is minimum (isothermal)
 - ☐ Fertility rate: implications of TFR < 2.1 for population structure
 - ☐ Probability: die with coloured faces; sequences GRRRR vs GRGRRR; comparison
 - ☐ Pouring problem: 12L, 8L, 5L vessels; minimum pourings to get 6L in 8L vessel

Complete Preparation Priority Map

Tier 1 — Master completely (do not skip any sub-topic)

1. **Real Analysis** — 9 major sub-areas, 50+ sub-topics
2. **Linear Algebra** — 9 major sub-areas, 45+ sub-topics
3. **Statistics & Probability** — 10 major sub-areas, 50+ sub-topics

Tier 2 — Cover fully; all sub-topics will appear

4. **Complex Analysis** — 7 major sub-areas, 30+ sub-topics
5. **Abstract Algebra** — 7 major sub-areas, 35+ sub-topics
6. **Ordinary Differential Equations** — 6 major sub-areas, 25+ sub-topics

Tier 3 — Selective mastery; pick your strongest topics

7. **PDEs** — 5 sub-areas; focus: d'Alembert, heat BVP, energy methods, Laplace max-principle
8. **Topology** — 5 sub-areas; focus: product topology, compactness (Tychonoff), separation axioms
9. **Functional Analysis** — 3 sub-areas; focus: ℓ^2 Hilbert space, compact sets in $C[0, 1]$
10. **Calculus of Variations & Mechanics** — 3 sub-areas; focus: Euler-Lagrange, isoperimetric, canonical transformations
11. **Numerical Methods** — 4 sub-areas; focus: N-R order (double zero $\Rightarrow p = 1$), Gauss-Seidel spectral radius, cubic splines
12. **Linear Programming** — 1 sub-area; straightforward: graphical method, simplex, read off optimal value

Part A — General Aptitude (always score 13+/15)

13. Arithmetic, ratio, proportion (8 Qs per paper — most predictable)
14. Logical reasoning: syllogisms, blood relations, coding (6 Qs — second most predictable)
15. Data interpretation: pie/bar graphs (3 Qs — read carefully, no tricks)

18 Guaranteed-Repeat Sub-topics (appear identically in BOTH 2025 papers): Decimal expansions (Q66) · Riemann integrable $[0, \infty)$ functions (Q67) · ODE $y'' + x^3 \sin(1/x)y = 0$ (Q68) · Product topology $\prod [0, c]$ (Q69) · $A^2 = I$, $\text{Tr}(A) = 0$ anticommutable (Q74) · $M/M/2$ queue $\lambda = 4, \mu = 1$ (Q75) · $M_n(\mathbb{C})$ inner product, $W_k = \{0\}$ (Q76) · Borrowing/troublesome (Q77) · Spanning set $\{u, v, w, x, y\}$ dimension (Q78) · Isoperimetric CoV $\int x^2 (y')^2$ (Q79) · Wishart $N_{12}(\mu, \Sigma)$, $n = 20$ (Q80) · $f_n = n(1 - nx)$ in $C[0, 1]$ (Q92) · $g(z) = f(z) - 1/z$ (Q93) · $H = \{(a_n) \in \ell^2 : |a_n| \leq 1/n\}$ (Q94) · Galois $G = S_p$ (Q108) · $T^2 = T$ projection span (Q109) · Non-abelian group $f(a, b) = ab$ (Q110) · $f(z) = \sin^2 z + \cos^2 |z|$ (Q120).